JMI Journal of Medical Imaging

RESEARCH PAPER

CoReg-Net: contrastive registration for structurally consistent one-shot medical image segmentation

Ziyi Xia,[a] Yongzhi Huang,[a] Feng Zhou,[a] Hongwei Zhang,[a] Yun Peng,[b] Yong Liu,[a] and Liyun Tu[a,*]

[a]Beijing University of Posts and Telecommunications, School of Artificial Intelligence, Beijing, China
[b]Beijing Children's Hospital, Capital Medical University, National Center for Children's Health, Department of Radiology, MOE Key Laboratory of Major Diseases in Children, Beijing, China

ABSTRACT. **Purpose:** In clinical scenarios where only a single labeled reference image is available for the target segmentation task, registration-based one-shot segmentation is a practical solution through anatomical label transfer. However, existing deep learning-based registration methods are often sensitive to appearance variations, which distort anatomical correspondence in the latent feature space and lead to structurally inconsistent deformation fields. This work aims to improve the structural consistency of deformable registration by explicitly aligning inter-subject anatomical representations.

Approach: We propose CoReg-Net, a contrastive registration framework that integrates an inter-subject contrastive learning (ISCL) module to enforce anatomically consistent feature representations by matching spatially corresponding regions across subjects. In parallel, a cascaded skip connection (CSC) mechanism is introduced to bridge the semantic gap between hierarchical encoder and decoder features, enabling more coherent reconstruction of fine-grained deformation details. These two components are jointly optimized to guide the network toward structurally meaningful and topologically regular deformation fields for robust label propagation.

Results: The proposed method demonstrates robust and consistent one-shot segmentation performance across diverse anatomical structures and imaging modalities on the IXI, OASIS, and BCV datasets. Compared with state-of-the-art registration-based baselines, CoReg-Net generates deformation fields with higher structural similarity to target images, confirming the effectiveness of contrastive feature alignment. Atlas replacement experiments further verify the robustness of the proposed method under atlas variation. Moreover, the framework exhibits adaptive deformability, preserving topological regularity in brain tissues while accurately capturing large non-rigid deformations in abdominal organs.

Conclusions: CoReg-Net shows that explicitly enforcing structural consistency through contrastive registration improves structural similarity and maintains segmentation accuracy. The proposed framework provides a reliable and generalizable solution for atlas-guided anatomical analysis across heterogeneous imaging modalities. Code is available at https://github.com/RelMono/CoRegNet.

© 2026 Society of Photo-Optical Instrumentation Engineers (SPIE) [DOI: 10.1117/1.JMI.13.4.044002]

Keywords: medical image segmentation; contrastive learning; image registration; one-shot learning; transformer

Paper 260041 received Jan. 22, 2026; revised Jun. 24, 2026; accepted Jul. 21, 2026; published Aug. 13, 2026.

*Address all correspondence to Liyun Tu, tuliyun@gmail.com

2329-4302/2026/$28.00 © 2026 SPIE

1 Introduction

Accurate delineation of anatomical structures from medical scans is central to quantitative analysis, treatment planning, and computer-aided diagnosis.[1] Over the past decade, deep learning-based segmentation models[2–6] have shown substantial improvements in performance across a wide range of medical imaging tasks, achieving accuracy comparable to expert annotations.[7] However, these fully supervised approaches rely on large-scale pixel-level annotations, which are difficult to obtain and limit their applicability in many clinical scenarios, especially for new anatomical targets or less common imaging settings.

To alleviate the reliance on labor-intensive manual annotation, one-shot segmentation aims to delineate target structures using only a single annotated reference image. Existing one-shot segmentation methods can be broadly categorized into prototype-based approaches[9,10] and augmentation-based strategies.[11–13] Although effective under limited annotation settings, these methods do not explicitly model spatial correspondence and therefore often struggle in applications that require anatomically consistent segmentation, such as atlas-guided medical image analysis. In contrast, registration-based one-shot segmentation methods (Fig. 1) estimate dense deformation fields to establish voxel-wise anatomical correspondence between a labeled reference image and a target image, enabling accurate label propagation.[14–16] Here, the one-shot setting refers to the scenario where only a single labeled reference image is available for the target task, and segmentations of unseen target images are obtained by propagating the reference annotation through deformable registration. Consequently, this paradigm has become a fundamental tool for quantitative anatomical analysis, atlas-based labeling, and image-guided surgical planning across diverse imaging modalities. Specifically, classical iterative methods, such as Demons[17] and SyN,[18] have been complemented by deep neural networks capable of predicting deformation fields in a single forward pass.[19,20] More recently, transformer-based registration

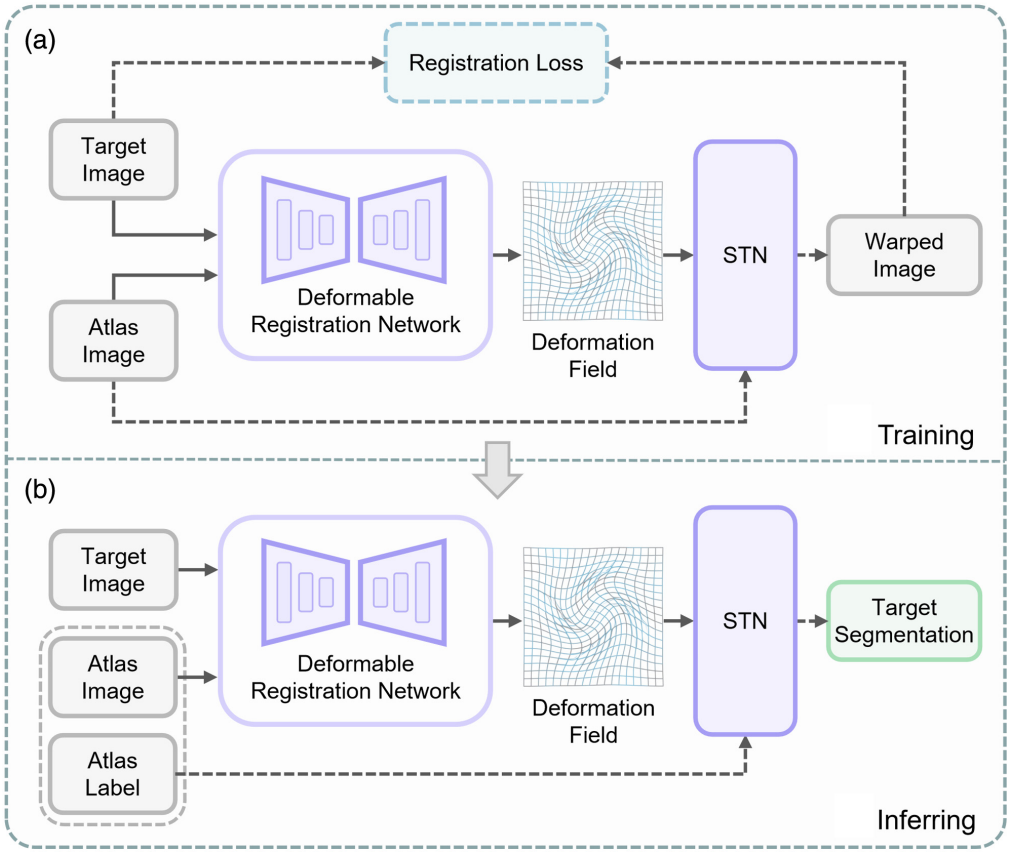


Fig. 1 Overview of registration-based one-shot segmentation. The deformable registration network is trained unsupervisedly to estimate a deformation field that aligns the atlas image with the target image. During inference, the generated field is applied to the atlas label using a spatial transformer network (STN)[8] to produce the target segmentation.

models, such as TransMorph,[21] have demonstrated strong performance by effectively modeling long-range anatomical dependencies. In parallel, advanced feature representation techniques, including wavelet-based modeling,[22] have further enhanced the characterization of fine-grained anatomical structures.

Despite these advances, when learning-based registration is employed for one-shot segmentation, two fundamental challenges remain unresolved. First, most registration networks rely on intensity-based similarity objectives and are therefore sensitive to acquisition protocols, contrast variations, and complex anatomical variability. These appearance- and shape-driven variations induce inconsistencies in latent representations across subjects, making it difficult to establish anatomically meaningful correspondences.[23,24] As a result, the estimated deformation fields may fail to achieve accurate anatomical alignment, directly affecting the reliability of label propagation in one-shot segmentation. Second, although Transformer backbones enhance global context modeling, they introduce a semantic gap between hierarchical encoder features and the spatially fine-grained decoder required for deformation prediction. This misalignment limits the reconstruction of subtle anatomical boundaries, particularly under multi-scale anatomical variability.[25–27]

To address these issues, we propose "Contrastive Registration Network" (CoReg-Net), a registration framework that enhances structural consistency in one-shot medical image segmentation through contrastive feature alignment and multiscale deformation modeling. CoReg-Net incorporates two key components: inter-subject contrastive learning (ISCL) and cascaded skip connection (CSC). ISCL is a contrastive learning strategy that enforces anatomically consistent feature representations by aligning spatially corresponding regions across subjects, thereby reducing sensitivity to appearance-driven variations and emphasizing stable anatomical structures. CSC is a cascading skip-connection mechanism that introduces progressive multiscale feature fusion, alleviating the semantic gap between encoder and decoder representations and enabling more accurate reconstruction of fine-grained anatomical details during deformation prediction. Integrated into a Swin Transformer-based registration backbone,[21] these components jointly improve the anatomical plausibility of the resulting deformation field, thereby supporting more reliable registration-based one-shot segmentation.

The main contributions of this study are as follows:

- We propose a registration framework that enhances structural consistency through contrastive alignment of anatomical features across subjects. The ISCL module reduces sensitivity to appearance variations and promotes anatomically aware latent representations.
- We introduce a cascaded skip connection mechanism that progressively aligns encoder and decoder features, mitigating semantic discrepancies and improving the representation of fine-grained anatomical structures in the deformation field.
- Experimental results on two brain MRI (IXI and OASIS) and one abdominal CT (BCV) datasets demonstrate statistically validated improvements over advanced baseline methods, indicating that CoReg-Net captures robust anatomical context and enables reliable registration-based one-shot segmentation.

2 Methods

2.1 Overview

Given an atlas image I_A with its corresponding label S_A and a target image I_T, CoReg-Net (Fig. 2) predicts a dense deformation field ϕ to align I_A with I_T. The network is trained in an unsupervised manner by maximizing an image similarity metric between the warped atlas image $I_A \circ \phi$ and the target image I_T. During inference, the estimated deformation field is applied to the atlas label through a spatial transformer network (STN),[8] yielding the final segmentation $S_T = S_A \circ \phi$.

CoReg-Net follows a U-Net-style encoder-decoder architecture. The encoder employs a Swin Transformer[21] as a hierarchical feature extractor, whereas the decoder is CNN-based to facilitate spatially precise deformation reconstruction. This backbone is incorporated with two architectural components: ISCL, applied to intermediate feature representations, and CSC, which enables progressive multiscale feature fusion along the decoder pathway.

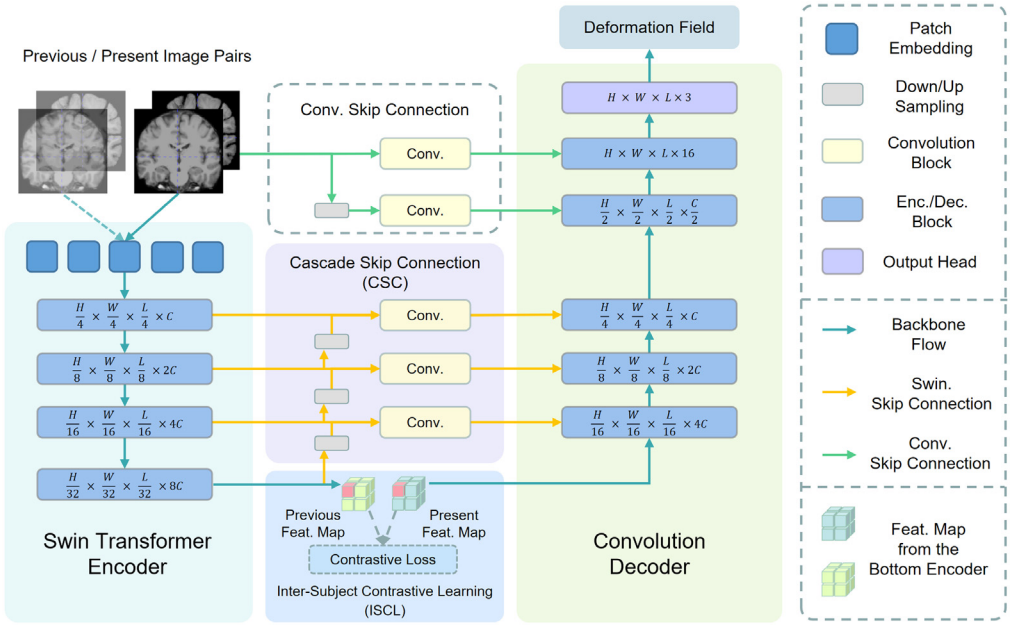


Fig. 2 Architecture of the proposed contrastive registration network (CoReg-Net). A Swin Transformer encoder extracts hierarchical features from the input image pair, and a CNN decoder progressively upsamples them to generate the deformation field. At the bottleneck, the ISCL module compares the current and stored feature maps from the previous iteration, partitioning them into patches to compute a contrastive loss that enforces anatomical consistency. The CSC mechanism replaces direct skip connections with progressive encoder-decoder feature fusion from deeper layers to reduce the semantic gap. Convolutional skip connections further inject raw image features (at original and 1/2 resolutions) into the upper decoder layers to preserve fine-grained details.

2.2 Inter-Subject Contrastive Learning for Anatomical Consistency

The auxiliary objective ISCL explicitly promotes anatomical consistency in the feature space across subjects. Rather than relying on appearance-driven similarity, ISCL encourages feature representations that are invariant to inter-subject intensity variations while remaining discriminative of anatomical location. This is achieved by aligning spatially corresponding regions across individuals, thereby introducing a structural inductive bias into the registration process. The ISCL module is applied to deep feature maps at the encoder bottleneck.

During training, ISCL operates on deep feature maps extracted from the encoder bottleneck. At each iteration, the feature map z_{curr} corresponding to the current input pair $(I_T, I_A)_{\text{curr}}$ is obtained, while the feature map z_{prev} from the previous iteration $(I_T, I_A)_{\text{prev}}$ is retrieved from a feature memory. To establish coarse anatomical correspondences across subjects, both 3D feature maps are spatially partitioned along their three axes into eight nonoverlapping patches of approximately equal size. Given the initial affine pre-alignment applied to all images, each spatially corresponding patch (e.g., top-left-front) is assumed to represent a consistent anatomical region across different subjects. Each patch is then aggregated into a single feature vector via average pooling, resulting in two sets of patch-level feature vectors, denoted as V_{curr} and V_{prev}.

Based on the patch sampling strategy, contrastive supervision is defined using spatial correspondence. For an anchor feature vector v_k^{curr} from the current set, the spatially corresponding vector v_k^{prev} from the previous set is treated as the positive pair, whereas all noncorresponding vectors v_j^{prev} where $j \neq k$ serve as negative pairs. This design encourages feature representations associated with the same coarse anatomical region to cluster together in the embedding space, while pushing apart representations from different regions. We use a contrastive loss to bring positive pairs closer and separate negative pairs in the embedding space, defined as:

$$\mathcal{L}_{\text{con}} = -\mathbb{E}\left[\log \frac{\exp(\text{sim}(v_k^{\text{curr}}, v_k^{\text{prev}})/\tau)}{\sum_{j=1, j\neq k}^{K} \exp(\text{sim}(v_k^{\text{curr}}, v_j^{\text{prev}})/\tau)}\right], \tag{1}$$

where $\mathrm{sim}(\cdot,\cdot)$ denotes cosine similarity and τ is a temperature hyperparameter. This loss explicitly guides the encoder to learn representations that are discriminative of anatomical structures.

2.3 Cascaded Skip Connection

The CSC mechanism is designed to reduce the semantic gap between encoder and decoder features, enabling more effective multiscale fusion for fine-grained anatomical structures. In standard U-Net architectures, shallow encoder features rich in local detail are directly concatenated with deeper decoder features that encode high-level semantics, creating an abrupt semantic shift that can hinder accurate reconstruction of subtle structures.[28] CSC alleviates this issue by introducing a progressive, hierarchical skip-fusion pathway that incrementally integrates multi-scale information into each decoder stage.

Formally, let e_l be the feature map from the l-th encoder stage. The cascaded feature map c_l, which is passed to the l-th decoder stage (after concatenation with the upsampled feature from the deeper decoder stage), is generated by a fusion process. This process starts at the bottleneck with $c_4 = e_4$. For shallower levels ($l = 1, 2, 3$), the cascaded feature map c_l is defined as

$$c_l = \mathrm{Conv}_l([e_l \| (w_l \cdot \mathrm{Upsample}(e_{l+1}))]), \tag{2}$$

where $[\cdot \,\|\, \cdot]$ denotes channel-wise concatenation, Conv_l is a 3D convolutional block, and w_l is a learnable scalar weight parameter.

This design introduces an adaptive weighting mechanism that allows the network to learn the relative importance of semantic features from deeper encoder levels at each stage of the decoder. Through weighted and progressive feature fusion, CSC ensures that each decoder stage receives rich contextual information, combining high-resolution spatial details from e_l with an adaptively scaled stream of semantic representations from deeper encoder layers. This design facilitates more accurate and stable reconstruction of the deformation field ϕ.

2.4 Loss Function

CoReg-Net is trained end-to-end by minimizing a composite loss function $\mathcal{L}$ for anatomically aware representations. To account for the distinct anatomical characteristics and deformation magnitudes of different imaging modalities, the similarity ($\mathcal{L}_{\mathrm{sim}}$) and regularization ($\mathcal{L}_{\mathrm{reg}}$) terms are specified separately for brain MRI and abdominal CT tasks. The overall objective function is defined as

$$\mathcal{L} = \lambda_{\mathrm{sim}}\mathcal{L}_{\mathrm{sim}} + \lambda_{\mathrm{reg}}\mathcal{L}_{\mathrm{reg}} + \lambda_{\mathrm{con}}\mathcal{L}_{\mathrm{con}}. \tag{3}$$

The parameters λ_{sim}, λ_{reg}, and λ_{con} modulate the balance among the three loss components.

For brain MRI registration, we used the local normalized cross-correlation (LNCC) loss for $\mathcal{L}_{\mathrm{sim}}$, which is robust to linear intensity variations and effectively captures local structural alignment. For each voxel p, LNCC is computed within a local window $w(p)$ of size n^3 ($n = 9$):

$$\mathcal{L}_{\mathrm{LNCC}}(p) = -\frac{\left(\sum_{q\in w(p)}(I_T(q) - \mu_T(p))((I_A \circ \phi)(q) - \mu_A(p))\right)^2}{\left[\sum_{q\in w(p)}(I_T(q) - \mu_T(p))^2\right]\left[\sum_{q\in w(p)}((I_A \circ \phi)(q) - \mu_A(p))^2\right]}, \tag{4}$$

where $\mu_T(p)$ and $\mu_A(p)$ denote the mean intensities of the target image I_T and the warped atlas image $I_A \circ \phi$ within the window $w(p)$, respectively.

To ensure that the predicted deformation field ϕ is smooth and physically plausible, the diffusion regularizer[19] is employed as $\mathcal{L}_{\mathrm{reg}}$ to penalize the spatial gradients of the deformation field:

$$\mathcal{L}_{\mathrm{diffusion}}(\phi) = \sum_p \|\nabla\phi(p)\|^2. \tag{5}$$

For the BCV dataset, global mutual information (GMI) is employed as $\mathcal{L}_{\mathrm{sim}}$ and the bending energy penalty is used as $\mathcal{L}_{\mathrm{reg}}$ to accommodate large non-rigid deformations and the complex intensity distribution of CT images. Specifically, GMI [Eq. (6)] captures nonlinear statistical dependencies between the source and target images, making it effective for multimodal or complex uni-modal registration scenarios. The bending energy penalty[29] [Eq. (7)] is adopted to

regularize the deformation field by penalizing its second-order derivatives, thereby discouraging folding and promoting smoother, fluid-like transformations.

$$\mathcal{L}_{\mathrm{GMI}} = -\sum_{i,j} p(i,j) \log \frac{p(i,j)}{p(i)p(j)}, \tag{6}$$

$$\mathcal{L}_{\mathrm{BE}}(\phi) = \sum_{p} \|\nabla^2 \phi(p)\|^2, \tag{7}$$

where $p(i)$ and $p(j)$ denote the marginal intensity distributions of the warped atlas and target images, respectively, and $p(i,j)$ is their joint distribution estimated via Parzen windowing; p denotes spatial locations in the deformation field.

3 Experiments and Results

3.1 Data Description

To comprehensively evaluate the robustness and generalization capability of CoReg-Net across different anatomical regions and imaging modalities, we conducted experiments on three publicly available datasets: IXI, OASIS, and BCV. This study used only publicly available, de-identified MRI and CT images from open sources and did not involve the collection of new imaging from human subjects. As the data were fully anonymized and accessed from existing datasets, institutional review board approval and informed consent were not required for this secondary analysis.

1. **IXI Dataset (Brain MRI):** This dataset contains 576 T1-weighted MRI scans collected from three hospitals with different scanners (Philips 3T, Philips 1.5T, GE 1.5T). The diversity in acquisition protocols provides a robust benchmark for scanner invariance. All scans are preprocessed using the standard FreeSurfer pipeline,[30] including motion correction, skull stripping, bias field correction, and resampling to an isotropic resolution of $1 \times 1 \times 1$ mm^3 within a unified grid of size $160 \times 192 \times 224$. Image intensities are normalized to the [0, 1] range. Following TransMorph,[21] the dataset is split into 403 subjects for training, 58 for validation, and 115 for testing (7:1:2). For the one-shot segmentation task, a single subject from the test set is randomly selected as the atlas, with an annotated label containing 30 anatomical structures. The remaining 114 test subjects serve as the evaluation set.
2. **OASIS Dataset (Brain MRI):** To further validate performance on brain anatomy, we utilized the Open Access Series of Imaging Studies (OASIS) dataset[31,32] from the Learn2Reg 2021 challenge.[33] Standard preprocessing steps were applied, including skull stripping, pre-registration, and intensity normalization to the [0, 1] range, resulting in a unified volume size of $160 \times 224 \times 192$. The dataset includes 414 T1-weighted MRI scans annotated with 35 anatomical structures, split into one atlas, 313 training, 50 validation, and 50 testing cases.
3. **BCV Dataset (Abdominal CT):** To challenge the model with large nonrigid deformations and inter-subject variability, we employed the Beyond the Cranial Vault (BCV) Abdomen dataset[34] from the Learn2Reg 2021 challenge.[33] It contains 50 CT scans with annotations for the liver, spleen, right kidney, and left kidney. Unlike brain MRI, abdominal CTs exhibit low soft-tissue contrast and significant anatomical variations. All scans were pre-registered and cropped to $192 \times 160 \times 192$.

3.2 Evaluation Metrics

We evaluate CoReg-Net on both deformable registration and registration-based one-shot segmentation. Registration quality is assessed from two complementary perspectives: (i) physical plausibility, quantified by the percentage of voxels with nonpositive Jacobian determinants ($|J| \leq 0\%$), denoted as the folding ratio, which indicates anatomically implausible folding, and (ii) perceptual similarity, measured using the structural similarity index measure (SSIM) between the warped atlas image ($I_A \circ \phi$) and the target image (I_T).

Segmentation performance is evaluated using two standard metrics: (i) volumetric overlap, measured by the Dice similarity coefficient (DSC), and (ii) boundary accuracy, assessed by the 95th percentile Hausdorff Distance (HD95). Segmentation results are reported on 17 composite anatomical structures obtained by merging bilateral regions (e.g., left and right lateral ventricles) into single categories, following common practice in registration-based one-shot segmentation. For consistency with prior work, HD95 and SSIM are computed using MONAI.[35]

3.3 Implementation Details

Our model was implemented in PyTorch[36] and trained on a single NVIDIA RTX 4090 GPU. We used the Adam optimizer with an initial learning rate of 1e-4, scheduled by polynomial decay. Training was conducted for 250 epochs with a batch size of 1. For brain MRI segmentation (IXI and OASIS), the loss weights were set to $\lambda_{\text{sim}} = 1$, $\lambda_{\text{reg}} = 1$, and $\lambda_{\text{con}} = 0.05$, with an ISCL temperature $\tau = 0.4$. For the BCV Abdominal CT dataset, the loss weights were set to $\lambda_{\text{sim}} = 1$, $\lambda_{\text{reg}} = 0.5$, and $\lambda_{\text{con}} = 0.05$, with an ISCL temperature $\tau = 0.4$. The final model was chosen based on the highest mean DSC on the validation set.

3.4 Baseline Methods

We compare CoReg-Net with several representative learning-based registration methods, including VoxelMorph,[19] CycleMorph,[37] MIDIR,[38] and nnFormer.[39] All baseline methods are evaluated under the same data split, preprocessing pipeline, and testing protocol. To ensure a fair comparison, all competing methods were evaluated under the same training protocol without method-specific hyperparameter optimization. For VoxelMorph, CycleMorph, and MIDIR, we follow the recommended settings reported in the original papers. As nnFormer was originally proposed as a U-Net-style segmentation architecture based on the Swin Transformer, we adapt it for deformable registration following the strategy of TransMorph,[21] using the same architectural backbone for deformation prediction.

For all baselines, we use the same training protocol as CoReg-Net, including the same optimizer, learning rate schedule, batch size, number of training epochs, and data preprocessing procedure. In addition, the same loss formulation is used across all methods so that the comparison focuses primarily on the effect of network architecture and method design, rather than differences in optimization settings.

3.5 Performance Evaluation on Brain MRI (IXI and OASIS)

3.5.1 *Performance evaluation on registration*

As shown in Table 1, CoReg-Net consistently achieves the highest SSIM scores (0.9402 on IXI and 0.9349 on OASIS), outperforming all baselines, including VoxelMorph and CycleMorph. This indicates that our method produces warped images that are perceptually closest to the target, effectively preserving fine anatomical textures and structural details.

Table 1 Quantitative evaluation of registration performance on brain MRI datasets.

	IXI Dataset		OASIS Dataset	
Method	SSIM ↑	$\lvert J\rvert \le 0\%$ ↓	SSIM ↑	$\lvert J\rvert \le 0\%$ ↓
VoxelMorph[19]	0.9192 ± 0.0103	0.5738 ± 0.2506	0.9185 ± 0.0053	0.5823 ± 0.0978
CycleMorph[37]	0.9267 ± 0.0107	0.6612 ± 0.2554	0.9254 ± 0.0049	0.6663 ± 0.0885
MIDIR[38]	0.8713 ± 0.0124	$\mathbf{0.0000 \pm 0.0000}$	0.8665 ± 0.0075	$\mathbf{0.0000 \pm 0.0000}$
nnFormer[39]	0.9243 ± 0.0110	0.6427 ± 0.2349	0.9232 ± 0.0052	0.6119 ± 0.1041
CoReg-Net (ours)	$\mathbf{0.9402 \pm 0.0108}$	0.5968 ± 0.2347	$\mathbf{0.9349 \pm 0.0046}$	0.5603 ± 0.0828

We report the structural similarity index (SSIM ↑) between the warped atlas image and target image (higher indicates better perceptual alignment) and the folding ratio ($\lvert J\rvert \le 0\%$, lower indicates smoother deformation).

Regarding physical plausibility, CoReg-Net maintains a low and stable folding ratio (0.5968% on IXI and 0.5603% on OASIS). On the OASIS dataset, our method yields a lower folding rate than VoxelMorph, CycleMorph, and nnFormer, indicating that CoReg-Net produces anatomically smooth deformations. The deformations in CoReg-Net avoid the over-smoothing behavior in MIDIR (as reflected in the lowest SSIM), which achieves near-zero folding by enforcing strict diffeomorphism via B-spline parameterization of stationary velocity fields but sacrifices registration accuracy. Meanwhile, it addresses the excessive folding problem frequently observed in CycleMorph. By balancing high perceptual fidelity with topological regularity, CoReg-Net provides a robust foundation for the superior segmentation accuracy observed in brain anatomy.

3.5.2 *Performance evaluation on one-shot segmentation*

On the IXI dataset (Table 2), CoReg-Net achieves the best overall performance and surpasses all competing baseline methods, achieving a mean DSC of 0.773 ± 0.112 and HD95 of 2.72 ± 0.96 mm. A structure-wise analysis shows that CoReg-Net attains the highest DSC in 15 of 17 structures.

On the OASIS dataset (Table 3), CoReg-Net achieves a mean DSC of 0.791 ± 0.152, outperforming nnFormer (0.786) and CycleMorph (0.783), while also attaining the lowest mean HD95 (2.52 mm). Across individual structures, our method ranks first in DSC for 11 of 19 structures, demonstrating robust generalization across different brain MRI cohorts.

A closer examination of specific structures further highlights these improvements. Across both datasets, the most substantial improvements are observed in anatomically complex and clinically relevant regions. On the IXI dataset, notable gains are observed in Putamen, Hippocampus, and Amygdala. Similarly, CoReg-Net excels in delineating fine-grained regions such as the cerebral white matter, cerebellar cortex, and brainstem on the OASIS dataset. These consistent observations suggest that the ISCL module effectively strengthens cross-subject anatomical consistency, while the CSC mechanism enables effective multiscale fusion that preserves fine-grained details. In contrast, simpler structures such as ventricles exhibit only modest improvements, as their regular geometry and well-defined boundaries already facilitate reliable registration.

To further assess whether these improvements are statistically reliable, we performed pairwise significance analysis between CoReg-Net and each competing method on the mean Dice and mean HD95 metrics across the IXI and OASIS datasets (Fig. 3). Specifically, for each dataset, per-case results were matched by case identity, and a paired Wilcoxon signed-rank test was conducted between CoReg-Net and each baseline method. The results show that the performance gains of CoReg-Net are statistically significant for most comparisons, confirming that the observed improvements are not merely due to random variation.

Overall, the results across two independent brain MRI datasets indicate that CoReg-Net achieves state-of-the-art registration-based one-shot segmentation performance, striking a favorable balance between volumetric overlap and boundary accuracy—particularly in anatomically intricate regions where accurate alignment is most challenging.

3.5.3 *Qualitative evaluation on registration and segmentation*

To complement the quantitative metrics, we provided qualitative comparisons on challenging subjects from both the IXI (Fig. 4) and OASIS (Fig. 5) datasets. Across both cohorts, CoReg-Net consistently generates segmentation masks that are most consistent with the ground truth, particularly in regions with intricate anatomical geometry.

On the IXI subject (Fig. 4), the differences are most evident in anatomically complex structures. In the top row, CoReg-Net preserves fine cortical folds and maintains clear separation between adjacent gyri, whereas baseline methods frequently confuse cortex and white matter, leading to blurred and anatomically inconsistent boundaries. A similar pattern appears in the yellow-boxed region of the bottom row, where CoReg-Net accurately delineates the putamen (light green) and its neighboring structures, while nnFormer and CycleMorph exhibit localized boundary leakage.

Comparable advantages are observed in the OASIS dataset (Fig. 5). CoReg-Net consistently yields sharp and coherent boundary delineations in fine-grained subcortical structures

Table 2 Quantitative comparison on the **IXI** dataset using DSC (↑) and HD95 (mm; ↓) over 17 aggregated anatomical structures (bilaterally merged).

	CWM	CC	LV	CbWM	CbC	Thal.	Caud.	Puta.	Pall.	3rd-V	4th-V	B-Stem	Hipp.	Amyg.	CSF	VDC	Ch-P	Average
Method										DSC ↑								
VoxelMorph[19]	0.807	0.725	0.794	0.791	0.856	0.843	0.746	0.782	0.740	0.709	0.755	0.898	0.763	0.692	0.610	0.784	0.341	0.743 ± 0.119
CycleMorph[37]	0.815	0.733	0.807	0.790	0.860	0.845	**0.764**	0.783	0.744	0.726	0.759	0.899	0.760	0.713	0.639	0.789	0.358	0.752 ± 0.115
MIDIR[38]	0.766	0.689	0.782	0.783	0.856	0.844	0.752	0.795	0.759	0.695	0.729	0.899	0.757	0.718	0.619	0.794	0.333	0.739 ± 0.120
nnFormer[39]	0.812	0.731	0.805	0.801	0.863	0.864	0.754	0.796	0.762	0.739	0.757	0.907	0.788	0.758	**0.658**	0.809	0.364	0.763 ± 0.114
CoReg-Net (ours)	**0.827**	**0.746**	**0.811**	**0.811**	**0.868**	**0.871**	0.763	**0.813**	**0.774**	**0.753**	**0.768**	**0.909**	**0.797**	**0.772**	0.652	**0.815**	**0.384**	**0.773 ± 0.112**
Method										HD95 (mm) ↓								
VoxelMorph[19]	2.75	2.32	4.52	3.11	3.67	2.67	2.35	2.37	2.61	2.93	3.27	2.58	2.75	2.93	3.04	2.27	5.96	3.06 ± 0.91
CycleMorph[37]	2.64	2.31	4.49	3.13	3.52	2.66	2.28	2.31	2.53	2.99	3.13	2.39	2.72	2.77	2.94	2.24	5.93	3.00 ± 0.91
MIDIR[38]	3.10	2.38	**3.94**	3.09	**2.96**	2.62	2.28	2.27	2.45	3.03	3.22	**2.07**	2.72	2.63	2.97	2.13	6.11	2.94 ± 0.92
nnFormer[39]	2.62	2.30	4.61	2.89	3.50	2.32	2.29	2.16	2.39	2.51	2.75	2.20	2.17	2.19	**2.84**	1.95	**5.60**	2.78 ± 0.94
CoReg-Net (ours)	**2.47**	**2.29**	4.34	**2.75**	3.60	**2.22**	**2.22**	**1.99**	**2.26**	**2.38**	**2.74**	2.29	**2.10**	**2.04**	2.91	**1.90**	5.69	**2.72 ± 0.96**

Mean ± standard deviation are reported, with the best results highlighted in bold. CWM: cerebral-white-matter, CC: cerebral-cortex, LV: lateral-ventricle, CbWM: cerebellum-white-matter, CbC: cerebellum-cortex, Thal.: thalamus-proper, Caud.: caudate, Puta.: putamen, Pall.: pallidum, 3rd-V: 3rd-ventricle, 4th-V: 4th-ventricle, B-Stem: brain-stem, Hipp.: Hippocampus, Amyg.: Amygdala, VDC: ventralDC, Ch-P: choroid-plexus.

Table 3 Quantitative comparison on the **OASIS** dataset using DSC (↑) and HD95 (mm; ↓) over 19 aggregated anatomical structures.

	CWM	CC	LV	ILV	CbWM	CbC	Thal.	Caud.	Puta.	Pall.	3rd-V	4th-V	B-Stem	Hipp.	Amyg.	SA	Ven	Vess	Ch-P	Average
Method	DSC ↑																			
VoxelMorph[19]	0.857	0.720	0.871	0.375	0.852	0.877	0.908	**0.861**	0.882	0.875	0.850	**0.801**	0.933	0.748	0.811	0.787	0.863	0.446	0.411	0.775 ± 0.166
CycleMorph[37]	0.863	0.734	0.870	0.402	0.858	0.881	0.911	**0.861**	0.884	0.875	**0.856**	**0.801**	0.934	0.748	0.827	0.797	0.868	0.497	0.412	0.783 ± 0.158
MIDIR[38]	0.803	0.669	0.849	0.308	0.826	0.860	0.905	0.831	0.866	0.866	0.831	0.782	0.927	0.737	0.815	0.781	0.858	0.395	0.385	0.752 ± 0.178
nnFormer[39]	0.864	0.728	**0.872**	**0.459**	0.857	0.883	**0.914**	0.858	0.886	0.875	0.853	0.778	0.936	**0.763**	**0.839**	0.797	0.872	0.484	**0.415**	0.786 ± 0.153
CoReg-Net (ours)	**0.873**	**0.744**	0.869	**0.459**	**0.867**	**0.888**	**0.914**	0.854	**0.895**	**0.887**	0.853	0.784	**0.938**	**0.763**	0.837	**0.809**	**0.877**	**0.503**	0.414	**0.791 ± 0.152**
Method	HD95 (mm) ↓																			
VoxelMorph[19]	2.56	1.62	3.57	6.05	3.34	3.02	1.62	1.47	1.75	1.36	1.38	**2.80**	1.91	2.94	1.92	1.59	1.48	3.56	**4.86**	2.57 ± 1.25
CycleMorph[37]	2.49	1.57	3.62	6.03	3.20	2.94	1.56	1.47	1.71	1.32	1.32	2.90	1.91	2.84	1.85	1.51	1.44	3.33	5.50	2.55 ± 1.33
MIDIR[38]	3.01	1.94	**3.44**	6.14	3.55	3.06	1.57	1.66	1.87	1.42	1.45	2.85	**1.59**	3.01	1.95	1.58	1.45	3.63	5.40	2.66 ± 1.32
nnFormer[39]	2.51	1.52	3.50	**5.56**	3.22	**2.75**	**1.53**	**1.44**	1.62	1.31	1.33	3.33	2.03	**2.74**	**1.72**	1.50	1.37	3.33	5.65	2.53 ± 1.30
CoReg-Net (ours)	**2.43**	**1.51**	3.60	5.67	**3.10**	2.78	**1.53**	1.52	**1.60**	**1.20**	**1.30**	3.17	1.97	2.78	1.74	**1.38**	**1.35**	**3.19**	6.01	**2.52 ± 1.36**

Mean ± standard deviation are reported, with the best results highlighted in bold. CWM: cerebral-white-matter, CC: cerebral-cortex, LV: lateral-ventricle, ILV: inf-horn-lat-vent, CbWM: cerebellum-white-matter, CbC: cerebellum-cortex, Thal.: thalamus-proper, Caud.: caudate, Puta.: putamen, Pall.: pallidum, 3rd-V: 3rd-ventricle, 4th-V: 4th-ventricle, B-Stem: brain-stem, Hipp.: Hippocampus, Amyg.: Amygdala, SA: septal-area, Ven: ventricle, Vess: vessels, Ch-P: choroid-plexus.

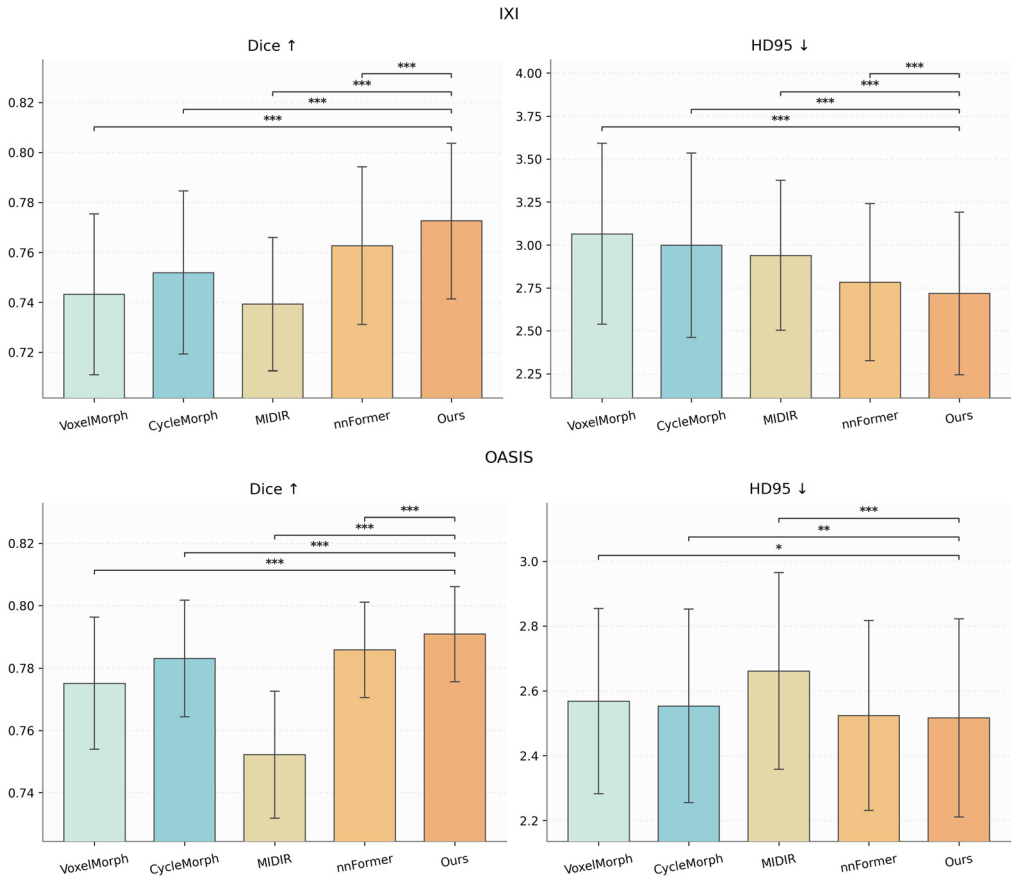


Fig. 3 Statistical significance analysis on the IXI and OASIS datasets. Mean Dice (higher is better) and mean HD95 (lower is better) are shown for all compared methods, with error bars indicating standard deviation. Statistical significance is evaluated between CoReg-Net and each competing method. Asterisks denote significance levels (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$).

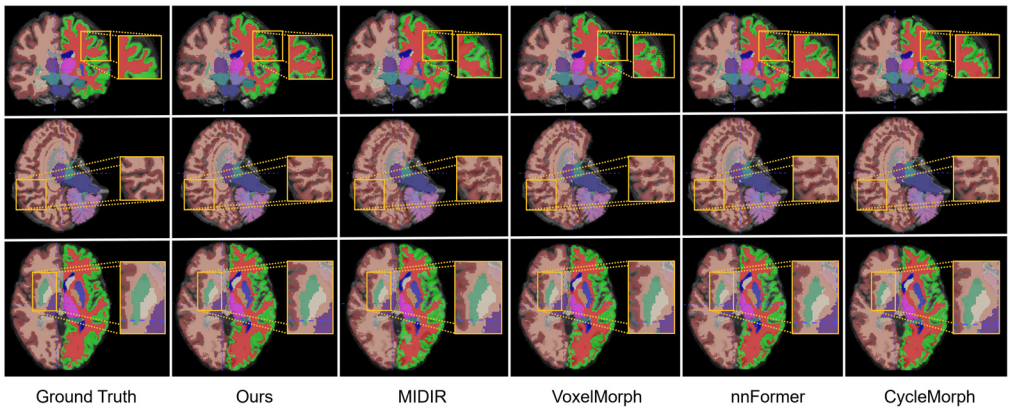


Fig. 4 Qualitative comparison of segmentation results in the IXI test set. The ground-truth segmentation is shown on the far left, followed by overlays from different methods. Yellow boxes highlight regions where our proposed CoReg-Net provides more accurate delineation, including the putamen and complex cortical folds.

(e.g., putamen and globus pallidus) and accurately captures the variability at the white matter-cortex interface. In contrast, baseline methods often produce over-smoothed segmentations or present local misalignment in these regions. Complementing these comparative evaluations, Fig. 6 provides a visual breakdown of the segmentation process on a representative IXI subject. The warped atlas demonstrates high structural consistency with the target image. This accurate

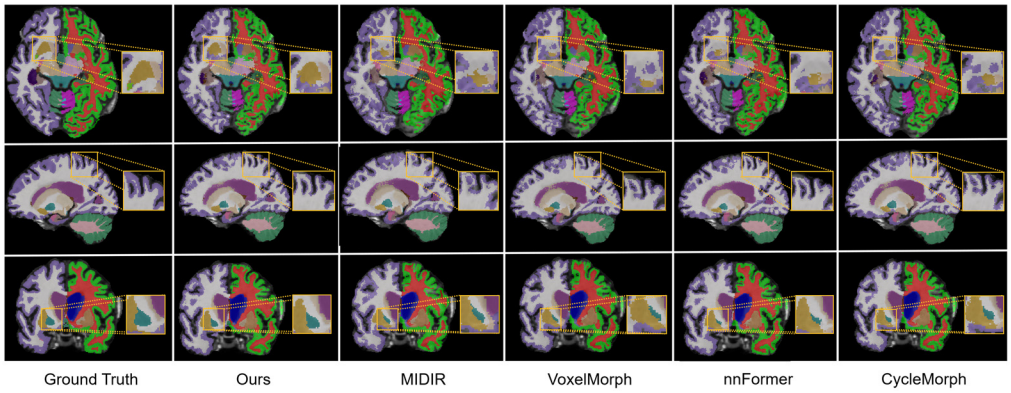


Fig. 5 Qualitative comparison of segmentation results in the OASIS test set. The ground-truth segmentation is shown on the far left, followed by overlays from different methods. Rows correspond to axial, sagittal, and coronal views. Yellow boxes highlight regions where our proposed CoReg-Net provides more accurate delineation.

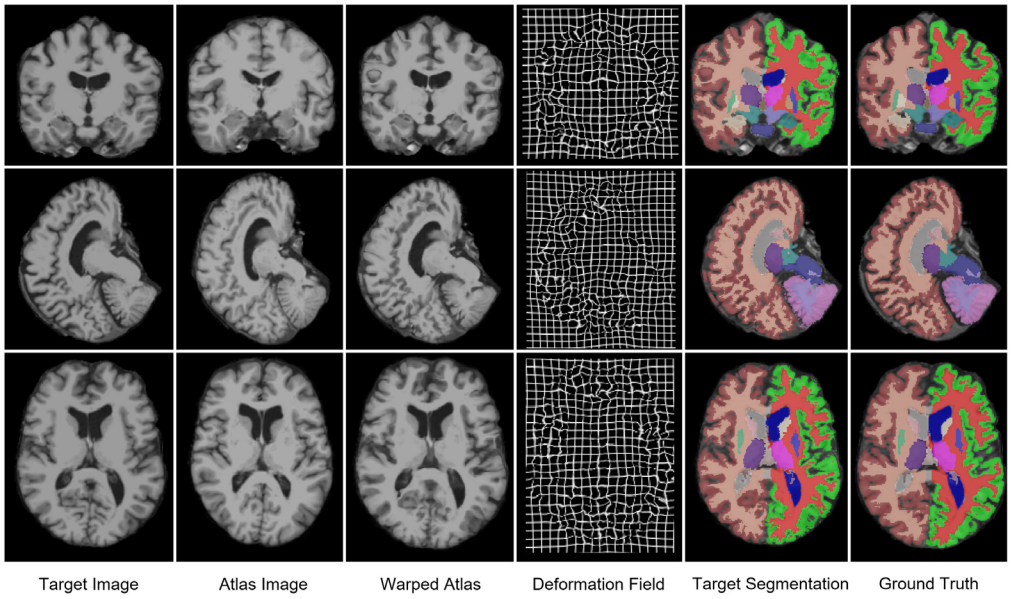


Fig. 6 Qualitative results of the full CoReg-Net pipeline on a representative subject in three different views. From left to right: the target image and atlas image; the warped atlas image obtained using the predicted deformation field; the resulting target segmentation produced by warping the atlas label; and the ground-truth of the target image.

alignment effectively translates to the label space, where the generated target segmentation closely follows the ground truth anatomy. These visualizations confirm that the enhanced feature representations contribute to more precise and robust anatomical delineations across diverse brain MRI datasets.

3.6 Performance Evaluation on Abdominal CT (BCV)

Beyond brain MRI, we further evaluate CoReg-Net on the abdominal CT-based BCV dataset, which poses substantially greater challenges due to low soft-tissue contrast and large non-rigid deformations inherent to abdominal organs. We benchmark CoReg-Net against the same baseline methods using both quantitative metrics (Table 4) and qualitative visualizations (Fig. 8) to assess its robustness and generalization capability in CT imaging.

As presented in Table 4, CoReg-Net demonstrates exceptional segmentation accuracy, achieving the highest mean DSC of 0.648, outperforming nnFormer (0.623) by approximately 0.025 and VoxelMorph (0.563) by 0.085. In structure-wise analysis, CoReg-Net ranks first across

Table 4 Quantitative evaluation on the BCV abdominal CT dataset.

Method	DSC ↑				Average DSC ↑	Average HD95 ↓	SSIM ↑	$\lvert J\rvert \leq 0\%$ ↓
	Liver	Spleen	R-Kidney	L-Kidney				
VoxelMorph[19]	0.771	0.546	0.497	0.439	0.563 ± 0.126	19.70 ± 0.54	0.4787 ± 0.0394	11.56 ± 1.66
CycleMorph[37]	0.796	0.649	0.504	0.452	0.600 ± 0.134	18.92 ± 0.71	0.5786 ± 0.0438	16.08 ± 3.07
MIDIR[38]	0.763	0.629	0.462	0.485	0.585 ± 0.121	$\mathbf{17.92 \pm 1.00}$	0.4938 ± 0.0385	$\mathbf{0.03 \pm 0.05}$
nnFormer[39]	0.797	0.676	0.525	0.496	0.623 ± 0.121	18.78 ± 1.49	0.4975 ± 0.0459	10.70 ± 2.54
CoReg-Net (ours)	**0.798**	**0.712**	**0.580**	**0.504**	$\mathbf{0.648 \pm 0.114}$	19.26 ± 2.60	$\mathbf{0.5925 \pm 0.0650}$	18.13 ± 1.28

We report DSC (↑) and HD95 (mm; ↓) for segmentation accuracy, along with the folding ratio ($|J| \leq 0\%$) and SSIM (↑) for registration quality. Mean ± standard deviation are reported. Best results are highlighted in bold.

all structures (liver, spleen, right kidney, and left kidney), indicating that the proposed contrastive feature alignment effectively captures large-scale semantic correspondences even under the low soft-tissue contrast of abdominal CT. In terms of registration quality, CoReg-Net also attains the highest SSIM (0.5925), reflecting improved alignment with the target image.

To complement the quantitative comparisons in Table 4, we further conducted pairwise significance analysis between CoReg-Net and each competing method on the per-case mean Dice and mean HD95 metrics (Fig. 7). The analysis indicates that the Dice improvements achieved by CoReg-Net are statistically significant across all baseline comparisons. By contrast, the HD95 results do not show the same consistent significance pattern, suggesting a trade-off between overlap accuracy and deformation smoothness.

To better understand this trade-off, we evaluated the predicted deformation fields using segmentation accuracy and topological regularity. As shown in Table 4, CoReg-Net exhibits a higher folding ratio (18.13%) and a slightly higher HD95 than MIDIR. This reflects an adaptive deformability: abdominal organs exhibit substantial inter-subject shape variability, often requiring more aggressive nonlinear transformations for accurate alignment. Although MIDIR maintains a near-zero folding ratio (≈0.03%) through strict diffeomorphic constraints, it struggles to model these large deformations, resulting in poor segmentation overlap (DSC 0.5848). In contrast,

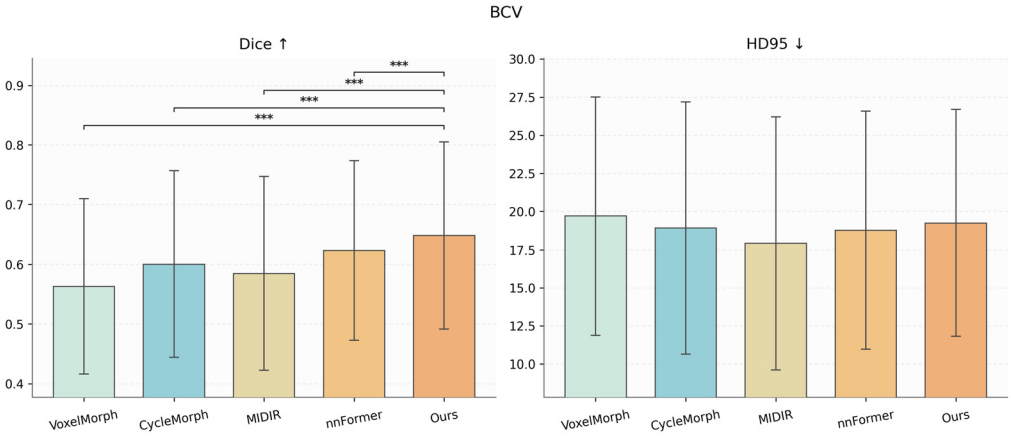


Fig. 7 Statistical significance analysis on the BCV dataset. Mean Dice (higher is better) and mean HD95 (lower is better) are shown for all compared methods, with error bars indicating standard deviation. Statistical significance is evaluated between CoReg-Net and each competing method. Significant improvements are observed for Dice over all baseline methods. Asterisks denote significance levels (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$).

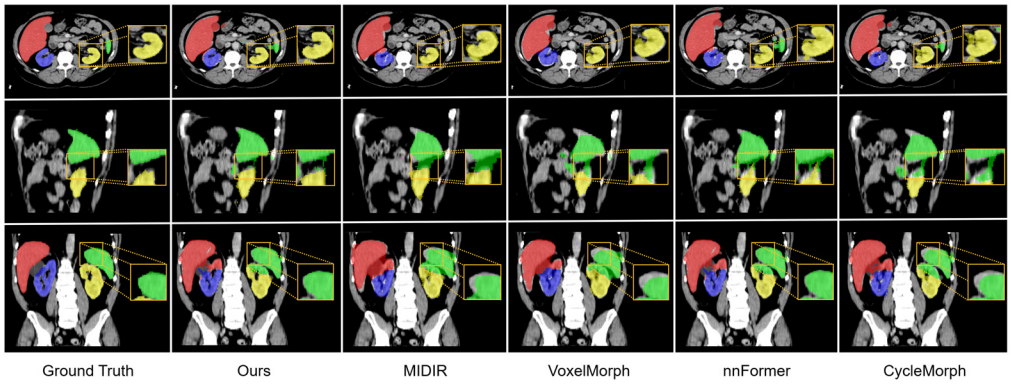


Fig. 8 Qualitative comparison of segmentation results on a representative subject from the BCV test set. Rows correspond to axial, sagittal, and coronal views. Yellow boxes highlight regions where our proposed CoReg-Net provides more accurate delineation.

CoReg-Net prioritizes anatomical overlap, effectively decoupling strict topological regularity for the flexibility required to align complex organ shapes. These findings suggest that, for one-shot abdominal segmentation, a controlled degree of deformation flexibility is essential for achieving high practical segmentation utility.

Figure 8 provides a visual comparison on a representative abdominal CT case characterized by substantial anatomical variability. The spleen and left kidney in the target image exhibit shapes and spatial configurations that differ markedly from those in the atlas. Baseline methods, particularly VoxelMorph and MIDIR, struggle to handle these large displacements, resulting in varying degrees of under-registration. As highlighted in the yellow boxes, these methods fail to stretch the atlas label sufficiently to cover the target organs, leading to significant false negatives. In contrast, CoReg-Net successfully captures the large-scale deformation, accurately warping the atlas left kidney (yellow) and spleen (green) to match the target boundaries. Although the underlying deformation field is less smooth, potentially causing minor boundary irregularities reflected in HD95, the resulting segmentation alignment is substantially improved.

3.7 Ablation Study

An ablation study was conducted on the IXI dataset to investigate the effect of the ISCL and CSC modules. As shown in Table 5, the full model achieves the most balanced overall performance across all evaluated metrics. Although the numerical gains in Dice and HD95 are relatively small, the inclusion of both modules leads to the highest SSIM and helps preserve deformation regularity. In particular, removing CSC results in a more noticeable drop in image similarity, whereas removing ISCL causes a larger increase in the non-positive Jacobian ratio. Overall, these findings suggest that ISCL and CSC play complementary roles in improving registration quality and structural plausibility.

Table 5 Ablation study of the proposed ISCL and CSC modules on the IXI dataset.

Variant	ISCL	CSC	Dice ↑	HD95 ↓	SSIM ↑	$\lvert J\rvert \leq 0\%$ ↓
Baseline	No	No	0.772 ± 0.111	2.72 ± 0.98	0.9388 ± 0.0113	0.5969 ± 0.2358
w/o ISCL	No	Yes	$\mathbf{0.773 \pm 0.111}$	2.75 ± 0.95	0.9399 ± 0.0109	0.6715 ± 0.2572
w/o CSC	Yes	No	0.771 ± 0.113	2.73 ± 0.97	0.9392 ± 0.0112	0.6221 ± 0.2381
CoReg-Net (ours)	Yes	Yes	0.773 ± 0.112	$\mathbf{2.72 \pm 0.96}$	$\mathbf{0.9402 \pm 0.0108}$	$\mathbf{0.5968 \pm 0.2347}$

We report segmentation accuracy using Dice (↑) and HD95 (mm; ↓), registration quality using SSIM (↑), and deformation regularity using the folding ratio ($\lvert J\rvert \leq 0\%$). Mean ± standard deviation are reported. Best results are highlighted in bold.

Table 6 Robustness analysis of CoReg-Net under different atlas choices on the IXI dataset.

Atlas Setting	Dice ↑	HD95 ↓	SSIM ↑	$\lvert J\rvert \leq 0\%$ ↓
Random IXI atlas	0.773 ± 0.112	2.72 ± 0.96	0.9402 ± 0.0108	0.5968 ± 0.2347
Highest-SSIM IXI atlas	0.7840 ± 0.1059	2.7654 ± 1.5170	0.9456 ± 0.0095	0.6740 ± 0.2699
Lowest-SSIM IXI atlas	0.7798 ± 0.1128	2.8718 ± 2.0535	0.9361 ± 0.0081	0.5656 ± 0.2145
External atlas	0.7470 ± 0.1160	3.6150 ± 1.9916	0.8850 ± 0.0179	1.1342 ± 0.2961

The test set is kept fixed, while the atlas is replaced by different in-domain and out-of-domain candidates. We report segmentation accuracy using Dice (↑) and HD95 (mm; ↓), registration quality using SSIM (↑), and deformation regularity using the folding ratio ($\lvert J\rvert \leq 0\%$). Mean ± standard deviation are reported.

3.8 Robustness to Atlas Selection

Because registration-based one-shot segmentation relies on the atlas to propagate the annotation, to evaluate the robustness of CoReg-Net to atlas selection, we conducted additional experiments on the IXI dataset while keeping the test set fixed and replacing the atlas with different candidates, including an in-domain atlas with the highest average SSIM to the training set, an in-domain atlas with the lowest average SSIM, and an external out-of-domain atlas. As shown in Table 6, CoReg-Net maintains relatively stable performance when different in-domain IXI atlases are used, with only limited variation in Dice, HD95, SSIM, and deformation regularity. In contrast, using the external atlas leads to a clear degradation across all metrics, indicating that although the proposed method is robust to atlas replacement within the same domain, its performance still depends on a reasonable anatomical and appearance match between the atlas and the target domain.

3.9 Hyperparameter Searching

We conducted an extended hyperparameter analysis on the IXI dataset by varying the smoothness weight $\lambda_{\text{reg}} \in \{0.5, 1, 1.5, 2\}$ and the contrastive loss weight $\lambda_{\text{con}} \in \{0.05, 0.1, 0.2\}$ while fixing $\lambda_{\text{sim}} = 1$. As shown in Fig. 9, the configuration $\lambda_{\text{reg}} = 1$ and $\lambda_{\text{con}} = 0.05$ achieves the best Dice score of 0.773, while the lowest HD95 of 2.68 is obtained at $\lambda_{\text{reg}} = 1.5$ and $\lambda_{\text{con}} = 0.1$. Overall, favorable performance is observed when λ_{reg} is in the range of 1 to 1.5 and λ_{con} is in the range of 0.05 to 0.1. We chose IXI for this sensitivity study because its relatively stable brain anatomy provides a cleaner setting for isolating the effects of individual loss weights. Accordingly, the observed parameter trends should be interpreted as empirical guidance under a controlled brain MRI setting rather than universally optimal choices across all datasets and modalities.

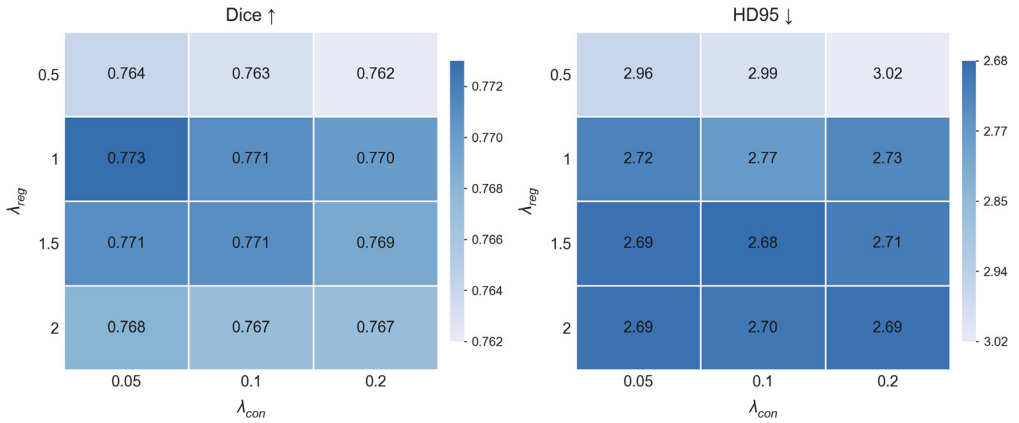


Fig. 9 Hyperparameter analysis of the loss weights λ_{reg} and λ_{con}. Dice and HD95 are evaluated across different combinations of the regularization and contrastive loss weights, as defined in Eq. (3).

4 Discussion and Conclusion

We introduced CoReg-Net, a deformable registration framework that enhances anatomically aware structural alignment in learning-based registration for one-shot segmentation. To address the challenges of appearance-driven latent inconsistency and the semantic gap between encoder and decoder representations, CoReg-Net explicitly enforces structural consistency at two complementary levels. In the latent feature space, ISCL encourages anatomically aligned feature representations across subjects, reducing reliance on intensity similarity alone. In the deformation space, CSC provides progressive multi-scale fusion to mitigate semantic discrepancies and enhance the reconstruction of fine-grained deformation details.

Empirically, CoReg-Net demonstrates consistently superior performance across multiple public datasets spanning brain MRI (IXI and OASIS) and abdominal CT (BCV), with statistical significance analyses further supporting its improvements over advanced baselines, indicating that the proposed design generalizes beyond a single anatomical domain or imaging modality. On brain MRI, CoReg-Net achieves competitive mean DSC (0.773 on IXI and 0.791 on OASIS) while simultaneously improving perceptual fidelity and reducing folding in the deformation field. These results suggest that enhanced anatomical consistency in the latent space translates into both more plausible deformations and more reliable label propagation for one-shot segmentation. On abdominal CT, where large nonrigid deformations and substantial inter-subject variability are common, CoReg-Net maintains strong performance (mean DSC 0.648), highlighting its adaptability to highly variable organ morphology. Qualitative visualizations further support these findings: CoReg-Net preserves fine cortical folding patterns in the brain while flexibly accommodating large-scale deformations in abdominal organs, illustrating improved structural alignment across different anatomical domains. Together, these results indicate that CoReg-Net provides a reliable, annotation-efficient solution for atlas-guided segmentation across diverse anatomical regions and imaging modalities. Additional atlas replacement experiments further show that CoReg-Net remains relatively stable under in-domain atlas variation, while performance degrades when the atlas is drawn from an out-of-domain distribution, highlighting the importance of domain-consistent anatomical priors in registration-based one-shot segmentation.

Despite these strengths, several limitations remain. First, very small or highly variable structures (e.g., the choroid plexus) remain challenging for registration-based methods due to weak contrast and irregular morphology. Although ISCL improves overall anatomical consistency, strategies incorporating structure priors or lightweight local refinement could further enhance performance for these anatomical regions. Second, like most learning-based registration frameworks, CoReg-Net relies on an initial alignment to establish coarse spatial correspondence. Extending the framework with coarse-to-fine alignment strategies or more transformation-invariant representations may improve robustness in cases of extreme inter-subject variability or poor initial alignment.

Disclosures

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Code and Data Availability

The used datasets are publicly available online through references cited in the manuscript. The source code is available at https://github.com/RelMono/CoRegNet.

Acknowledgments

This work was supported in part by the Beijing Natural Science Foundation-Haidian Original Innovation Joint Fund (Grant No. L232138), and the National Natural Science Foundation of China (Grant No. 62201091).

References

1. S. Minaee et al., "Image segmentation using deep learning: a survey," *IEEE Trans. Pattern Anal. Mach. Intell.* **44**(7), 3523–3542 (2021).

2. O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," *Lect. Notes Comput. Sci.* **9351**, 234–241 (2015).
3. F. Isensee et al., "nnU-net: a self-configuring method for deep learning-based biomedical image segmentation," *Nat. Methods* **18**(2), 203–211 (2021).
4. J. Chen et al., "Transunet: transformers make strong encoders for medical image segmentation," arXiv:2102.04306 (2021).
5. Y. Gao, M. Zhou, and D. N. Metaxas, "Utnet: a hybrid transformer architecture for medical image segmentation," *Lect. Notes Comput. Sci.* **12903**, 61–71 (2021).
6. A. Hatamizadeh et al., "Swin UNETR: swin transformers for semantic segmentation of brain tumors in MRI images," *Lect. Notes Comput. Sci.* **12962**, 272–284 (2021).
7. R. Azad et al., "Medical image segmentation review: the success of u-net," *IEEE Trans. Pattern Anal. Mach. Intell.* **46**, 10076–10095 (2024).
8. M. Jaderberg et al., "Spatial transformer networks," in *Adv. Neural Inf. Process. Syst.*, Vol. 28 (2015).
9. A. Balasundaram et al., "A foreground prototype-based one-shot segmentation of brain tumors," *Diagnostics* **13**(7), 1282 (2023).
10. S. Tang et al., "Few-shot medical image segmentation with high-fidelity prototypes," *Med. Image Anal.* **100**, 103412 (2025).
11. W. Liu et al., "One-shot segmentation of novel white matter tracts via extensive data augmentation," *Lect. Notes Comput. Sci.* **13431**, 133–142 (2022).
12. W. Liu et al., "One-shot segmentation of novel white matter tracts via extensive data augmentation and adaptive knowledge transfer," *Med. Image Anal.* **90**, 102968 (2023).
13. L. Zhang et al., "One-shot neuroanatomy segmentation through online data augmentation and confidence aware pseudo label," *Med. Image Anal.* **95**, 103182 (2024).
14. A. Sotiras, C. Davatzikos, and N. Paragios, "Deformable medical image registration: a survey," *IEEE Trans. Med. Imaging* **32**(7), 1153–1190 (2013).
15. S. Wang et al., "LT-Net: label transfer by learning reversible voxel-wise correspondence for one-shot medical image segmentation," in *Proc. IEEE/CVF Conf. on Comput. Vision and Pattern Recognit.*, pp. 9162–9171 (2020).
16. L. W. Remedios et al., "Influence of early through late fusion on pancreas segmentation from imperfectly registered multimodal magnetic resonance imaging," *J. Med. Imaging* **12**(2), 024008 (2025).
17. T. Vercauteren et al., "Diffeomorphic demons: efficient non-parametric image registration," *NeuroImage* **45**(1), S61–S72 (2009).
18. B. B. Avants et al., "Symmetric diffeomorphic image registration with cross-correlation: evaluating automated labeling of elderly and neurodegenerative brain," *Med. Image Anal.* **12**(1), 26–41 (2008).
19. G. Balakrishnan et al., "Voxelmorph: a learning framework for deformable medical image registration," *IEEE Trans. Med. Imaging* **38**(8), 1788–1800 (2019).
20. Y. Fu et al., "Deep learning in medical image registration: a review," *Phys. Med. Biol.* **65**(20), 20TR01 (2020).
21. J. Chen et al., "Transmorph: transformer for unsupervised medical image registration," *Med. Image Anal.* **82**, 102615 (2022).
22. X. Zhang et al., "Wavelet-guided multi-scale convnext for unsupervised medical image registration," *Bioengineering* **12**(4), 406 (2025).
23. L. Liu, A. I. Aviles-Rivero, and C.-B. Schönlieb, "Contrastive registration for unsupervised medical image segmentation," *IEEE Trans. Neural Netw. Learn. Syst.* **36**(1), 147–159 (2023).
24. L. Liu et al., "PC-swinmorph: patch representation for unsupervised medical image registration and segmentation," arXiv: 2203.05684 (2022).
25. Y. Zhang et al., "Deformable image registration with strategic integration pyramid framework for brain MRI," *Magn. Reson. Imaging* **120**, 110386 (2025).
26. M. Kang et al., "Dual-stream pyramid registration network," *Med. Image Anal.* **78**, 102379 (2022).
27. T. C. Mok and A. C. Chung, "Large deformation diffeomorphic image registration with laplacian pyramid networks," *Lect. Notes Comput. Sci.* **12263**, 211–221 (2020).
28. Z. Zhou et al., "UNet++: redesigning skip connections to exploit multiscale features in image segmentation," *IEEE Trans. Med. Imaging* **39**(6), 1856–1867 (2019).
29. D. Rueckert et al., "Nonrigid registration using free-form deformations: application to breast mr images," *IEEE Trans. Med. Imaging* **18**(8), 712–721 (1999).
30. B. Fischl, "Freesurfer," *NeuroImage* **62**(2), 774–781 (2012).
31. D. S. Marcus et al., "Open access series of imaging studies (oasis): cross-sectional mri data in young, middle aged, nondemented, and demented older adults," *J. Cogn. Neurosci.* **19**(9), 1498–1507 (2007).
32. A. Hoopes et al., "HyperMorph: amortized hyperparameter learning for image registration," *Lect. Notes Comput. Sci.* **12729**, 3–17 (2021).
33. A. Hering et al., "Learn2Reg: comprehensive multi-task medical image registration challenge, dataset and evaluation in the era of deep learning," *IEEE Trans. Med. Imaging* **42**(3), 697–712 (2022).

34. Z. Xu et al., "Evaluation of six registration methods for the human abdomen on clinically acquired CT," *IEEE Trans. Biomed. Eng.* **63**(8), 1563–1572 (2016).
35. M. J. Cardoso et al., "Monai: an open-source framework for deep learning in healthcare," arXiv:2211.02701 (2022).
36. A. Paszke et al., "Pytorch: an imperative style, high-performance deep learning library," in *Adv. Neural Inf. Process. Syst.*, Vol. 32 (2019).
37. B. Kim et al., "Cyclemorph: cycle consistent unsupervised deformable image registration," *Med. Image Anal.* **71**, 102036 (2021).
38. H. Qiu et al., "Learning diffeomorphic and modality-invariant registration using b-splines," in *Med. Imaging with Deep Learn.*, pp. 645–664 (2021).
39. H.-Y. Zhou et al., "nnFormer: interleaved transformer for volumetric segmentation," arXiv:2109.03201 (2021).

Biographies of the authors are not available.